A complete technical review of every SEO
infrastructure deployed on snbdhost.com — w
and how it benefits your search ranking

12	Pages Optimised
7	FAQ Schemas Added
3	Structured Data Types
100%	Crawlers Covered

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SECTION 1

Executive Summary

This report documents the complete SEO infrastructure and blog publishing system implemented on the SNBD HOST website (snbdhost.com) — a React 19 single-page application built with Vite 8 and React Router DOM 7. Prior to this implementation, the website had zero per-page SEO configuration: all 12 pages shared a single global title and description set in the root HTML file, with no sitemap, no robots.txt, no structured data, and no blog section.

The implementation addresses every critical pre-launch SEO requirement and adds a fully functional blog publishing system. The work spans both frontend (React) and backend (Node.js + Express) layers, and provides a complete content management workflow from writing a post to it appearing in Google search results.

Key outcomes of this implementation

- Every page now has a unique, keyword-optimised title and meta description
- Social sharing (Facebook, Twitter, LinkedIn, WhatsApp) now shows rich previews for all pages and blog posts
- Google and other search engines can discover all pages via a dynamic sitemap.xml
- 7 pages now qualify for Google FAQ rich snippets — increasing click-through rates
- A fully functional blog with admin panel is ready to publish SEO content
- The website is now ready to be submitted to Google Search Console

SECTION 2

Before & After: What Changed

2.1 Critical SEO Gaps That Existed

The following table shows the state of the website before this implementation. Every item marked Critical would directly prevent proper search engine indexing and ranking.

Feature	Status	Impact
Per-page title tags	MISSING	All 12 pages shared one generic title
Per-page meta description	MISSING	No unique descriptions for any page
Open Graph (og:*) tags	MISSING	Social shares showed no preview
Twitter Card tags	MISSING	Twitter/X showed no card
robots.txt	MISSING	Crawlers had no directives
sitemap.xml	MISSING	Pages could not be submitted to Google
JSON-LD Structured Data	MISSING	No rich snippets possible
Canonical URLs	MISSING	Duplicate content risk
Blog section	MISSING	Nav link pointed to #
CMS / Admin panel	MISSING	No way to publish content

2.2 What Was Fixed and Added

Feature Added	What it Does	Location
react-helmet-async	Unique SEO meta per page	All 12 pages
Open Graph + Twitter Card	Rich social previews on every share	All 12 pages
JSON-LD — Organisation	Google Knowledge Panel eligibility	All pages (global)
JSON-LD — FAQPage	FAQ rich snippets in Google search results	7 pages
JSON-LD — BlogPosting	Article rich results for blog posts	Every blog post
robots.txt	Crawler directives + sitemap reference	public/ (static)
sitemap.xml (static)	13 static routes discoverable by Google	public/ (static)
sitemap.cjs (dynamic)	Regenerates with blog post URLs on every build	scripts/

Bot meta injection	Social bots get correct OG tags server-side	Express server
Blog backend API	Create, edit, publish, delete posts via JWT	server/
Blog admin panel	Markdown editor + SEO sidebar at /admin	src/pages/admin/
Blog listing /blog	Paginated, styled post grid	src/pages/blog/
Blog post /blog/:slug	Rendered Markdown, tags, social share	src/pages/blog/
Header Blog link	Fixed from href='#' to Link to='/blog'	src/components/

SECTION 3

Per-Page SEO Meta Tags

This is the single most impactful change for search engine rankings. Previously, Google saw the same title and description for every page — making it impossible to rank each page for its own keywords.

3.1 How react-helmet-async Works

React is a Single Page Application (SPA) — the browser renders the page using JavaScript. This means the HTML document served to the browser is nearly empty until JavaScript runs. The react-helmet-async library solves this by dynamically updating the document section whenever a React component renders, injecting the correct meta tags for the current route.

Google's crawler executes JavaScript and picks up these dynamically-set tags. For social bots (Facebook, Twitter, WhatsApp) which do not run JavaScript, the Express server performs server-side meta injection — covered in Section 7.

3.2 All 12 Pages Optimised

Each page now has a unique, keyword-targeted title and description:

Route	Page Title (keyword focus)	Description focus
/	Bangladesh's #1 Web Hosting Provider	NVMe SSD, BDIX, 99.9% uptime, 24/7 support
/hosting	Shared Web Hosting Bangladesh — NVMe SSD	LiteSpeed, cPanel, free SSL from ■99/mo
/reseller-hosting	Reseller Hosting Bangladesh — White Label	WHM, CloudLinux, unlimited cPanel accounts
/domain	Domain Registration Bangladesh — .com .bd .xyz	50+ TLDs, free WHOIS privacy
/vps-server	VPS Server Bangladesh — KVM Linux VPS	AMD EPYC, NVMe SSD, full root access
/bdix-servers	BDIX Server Bangladesh — Ultra-Fast Local VPS	Sub-ms ping Dhaka, from ■500/mo
/openclaw	OpenClaw — Deploy AI Agent in 60 Seconds	Private, always-on AI agent platform
/n8n-automation	n8n Automation Hosting Bangladesh	Managed, instant setup from ■250/mo
/offers	Web Hosting Offers & Discount Codes	Promo codes, save up to 75%
/support	Customer Support — SNBD HOST Help Center	24/7, under 15-min response
/privacy	Privacy Policy — SNBD HOST	GDPR & CCPA compliant
/terms	Terms of Service — SNBD HOST	Bangladesh law jurisdiction

3.3 Open Graph & Twitter Card Tags

Every page now sets the full set of social sharing meta tags. This means when anyone shares a link to your website on Facebook, LinkedIn, Twitter/X, WhatsApp, or Telegram, they will see a rich preview card with your logo, the page title, and description — instead of a plain URL with no preview.

og:title	Page-specific title shown in the social card preview
og:description	Concise description shown under the title
og:image	SNBD HOST logo (or blog post's featured image)
og:url	Canonical URL for the page — prevents duplicate shares
og:type	'website' for all pages, 'article' for blog posts
og:site_name	Always 'SNBD HOST' — consistent brand name
twitter:card	summary_large_image — shows a large image preview on X
twitter:title	Optimised title for the Twitter card
twitter:description	Concise description for the Twitter card
twitter:image	Large preview image for the Twitter card

SECTION 4

Structured Data (JSON-LD)

Structured data tells Google exactly what your content means — not just the words, but the type of entity (Organisation, FAQ, Article). When Google understands your content structurally, it can show enhanced search results called 'rich snippets', which appear larger in search results and have higher click-through rates.

4.1 Organisation Schema

The Organisation schema is injected globally on every page. It tells Google who SNBD HOST is as a business entity — enabling your brand name and logo to appear in Google's Knowledge Panel when users search directly for 'SNBD HOST'. It also establishes E-E-A-T signals (Experience, Expertise, Authoritativeness, Trustworthiness) that Google uses to assess content quality.

4.2 FAQPage Schema — 7 Pages

Seven product pages already had FAQ sections built into their JSX components. The implementation maps each existing FAQ array into the schema.org FAQPage format and injects it as a JSON-LD script tag. No new FAQ content was needed — existing content was structured.

What FAQPage schema gives you in Google

- FAQ dropdowns appear directly in the Google search result — without the user clicking your link
- Your result occupies 3–5x more vertical space on the search results page
- Users can see your FAQ answers before visiting the site, establishing trust
- Higher visibility leads to significantly higher click-through rates (typically 20–30% uplift)
- Works on both desktop and mobile Google search results

Page	FAQ Count	Topics Covered
/vps-server	4 FAQs	Root access, OS support, dedicated resources, backups
/bdix-servers	4 FAQs	BDIX explanation, root access, upgrades, control panels
/n8n-automation	5 FAQs	Social media, abandoned cart, coding, vs Zapier, errors
/openclaw	4 FAQs	Technical skills, local LLM, data security, scaling
/offers	5 FAQs	Promo codes, new client discounts, max discount, coupons
/reseller-hosting	4 FAQs	White-label, overselling, migration, client isolation
/support	9 FAQs	Server access, order status, connection, password, billing

4.3 BlogPosting Schema

Every blog post page dynamically generates a BlogPosting JSON-LD schema using the post's metadata from the database. This includes the headline, author, publisher, datePublished, dateModified, and image fields. This makes blog posts eligible for Google's Article rich results and helps establish the post's publication date in search results.

SECTION 5

Static SEO Files

5.1 robots.txt

The robots.txt file gives instructions to all web crawlers about which parts of the site they are and are not allowed to index. Without it, crawlers operate without guidance and may waste crawl budget on pages like /admin that should never appear in search results.

User-agent: *	Applies to all crawlers (Google, Bing, Yandex, DuckDuckGo, etc.)
Allow: /	All public pages are crawlable
Disallow: /admin	The admin panel will never appear in any search engine
Disallow: /api/	API endpoints are excluded from indexing
Sitemap: reference	Points crawlers directly to the sitemap URL

5.2 sitemap.xml

A sitemap is a file that lists every URL on your website and tells Google how frequently each page is updated and how important it is relative to other pages. Without a sitemap, Google has to discover pages by following links — which can take weeks or miss pages entirely.

Two sitemap mechanisms are in place:

File	Type	Coverage	Purpose
public/sitemap.xml	Static	13 routes	Always available — served immediately
scripts/generate-sitemap.cjs	Dynamic	13 + all blog posts	Run at build time — includes every published blog post URL

URL	Priority	Change Frequency
/	1.0	weekly
/hosting	0.9	weekly
/reseller-hosting	0.9	weekly
/vps-server	0.9	weekly
/domain	0.8	weekly
/bdix-servers	0.8	weekly
/offers	0.8	daily

/blog	0.8	daily
/openclaw	0.7	monthly
/n8n-automation	0.7	monthly
/support	0.6	monthly
/privacy	0.3	yearly
/terms	0.3	yearly
/blog/:slug	0.7	monthly

SECTION 6

Blog Publishing System

A blog is the most effective long-term SEO tool for a hosting company. Every article you publish is a new page that can rank for a keyword, attract new visitors, and build the domain's authority. The blog system is fully custom-built, requiring no third-party CMS subscriptions.

6.1 Backend API

The blog backend is a Node.js + Express server running on port 3001 with a SQLite database. SQLite was chosen for its zero-setup simplicity — the entire database is a single file (data/blog.db) that can be backed up with a single copy command. The server is designed to be run alongside Nginx on the production VPS.

Endpoint	Auth	Description
POST /api/auth/login	Public	Email + password → JWT token (7-day expiry)
GET /api/posts	Public	Paginated published posts. ?status=all for admin
GET /api/posts/:slug	Public	Single post by URL slug (published only)
POST /api/posts	JWT Required	Create new post — auto-generates slug from title
PUT /api/posts/:id	JWT Required	Update post content, status, or SEO fields
DELETE /api/posts/:id	JWT Required	Permanently delete a post
GET /health	Public	Server health check for monitoring

Blog Post Data Structure

title	The post headline — also used to auto-generate the URL slug
slug	URL-safe version of the title, e.g. how-to-set-up-wordpress
excerpt	Short summary shown in the blog listing cards
content	Full post body written in Markdown
author	Author name shown on the post page
category	Category label shown as a badge (e.g. WordPress, VPS)
tags	Comma-separated tags stored as JSON, shown at post bottom
featured_image_url	URL to the post's hero image
status	draft or published — only published posts appear on /blog
meta_title	Custom SEO title (overrides post title in Google)

meta_description	Custom SEO description (150-160 chars for Google snippets)
og_image	Custom Open Graph image for social sharing previews
published_at	Timestamp set automatically when first published

6.2 Admin Panel

A fully functional admin interface is available at /admin/login. After authenticating with the admin credentials, you can manage all blog content through a clean dark-themed dashboard.

Posts dashboard /admin	Lists all posts (drafts + published) with status badges, dates, and quick actions
Post editor /admin/posts/new	Full Markdown editor with live preview tab, SEO sidebar, and status toggle
Edit existing post	Load any post's content into the editor for changes
Publish / Unpublish	One-click toggle between draft and published states
Delete post	Permanently remove a post from the database
SEO sidebar	Per-post meta_title, meta_description (with char counter), og_image fields
Markdown support	Full GitHub-Flavoured Markdown with live preview: headers, bold, code, lists, links
Auth protection	JWT stored in localStorage — expired or missing tokens redirect to login

6.3 Blog Frontend

/blog listing page	3-column responsive grid of post cards. Each card shows category badge, title, excerpt, author, date, and reading time estimate
Post cards	No-image placeholder (newspaper icon) if no featured image. Featured image lazy-loaded if set
Pagination	Previous/Next page buttons updating the URL (?page=2) — supports unlimited posts
/blog/:slug post page	Full Markdown rendered via marked + DOMPurify sanitisation. No XSS vulnerabilities
BlogPosting JSON-LD	Injected per-post for Google Article rich results
Tags display	Hashtag badges at the bottom of each post
Social share buttons	One-click share to X, Facebook, and LinkedIn using static URL parameters
Related posts	3 related posts from the same category shown at the bottom
Back link	← Back to Blog link at the top for easy navigation
SEO per post	react-helmet-async sets title, description, canonical, og:*, and twitter:* from post's own meta fields

SECTION 7

Bot-Aware Meta Tag Injection

React SPAs present a fundamental SEO challenge: the browser must run JavaScript to render the page. Google's crawler does run JavaScript, but social bots (Facebook, Twitter, WhatsApp, Telegram, LinkedIn, Discord) do not. Without a server-side solution, sharing any page link on social media would show no preview at all.

The Express blog server solves this with bot detection middleware. Every non-API request to the server is checked against a list of known bot User-Agent strings. If a bot is detected, the server reads index.html, injects the correct meta tags for that specific URL into the HTML, and serves the enriched document. Human visitors receive the standard index.html and the React app runs normally.

Bot User-Agent	Platform
facebookexternalhit	Facebook / Instagram link previews
twitterbot	Twitter / X card previews
linkedinbot	LinkedIn post previews
whatsapp	WhatsApp link previews
telegram	Telegram URL previews
discordbot	Discord embed previews
slackbot	Slack unfurl previews
applebot	Apple Spotlight and Safari suggestions
googlebot	Uses react-helmet-async (runs JS)
bingbot	Uses react-helmet-async (runs JS)

For blog post URLs (/blog/:slug), the server also queries the SQLite database to retrieve the post's meta_title, meta_description, og_image, and featured_image_url — ensuring each individual blog post gets its own rich social preview, not just a generic site preview.

SECTION 8

SEO Benefits & Impact Analysis

This section explains in plain terms how each implemented feature directly benefits snbdhost.com's search engine performance, traffic, and audience reach.

Per-page meta tags

Impact timeline: Short-term: Immediate impact

- Each page can now rank for its own specific keywords rather than competing with each other
- Google will show the correct page title in search results (e.g. 'VPS Server Bangladesh' for /vps-server)
- Unique descriptions act as ad copy — well-written descriptions increase click-through rates by 5–15%
- This is the foundational requirement for any SEO effort — nothing else works without it

sitemap.xml submission

Impact timeline: Short-term: Within 1–2 weeks

- After submitting to Google Search Console, Google will prioritise crawling all 13 pages
- New blog posts appear in Google's index typically within hours of being published
- The dynamic sitemap automatically includes new blog posts after every build

FAQPage JSON-LD (7 pages)

Impact timeline: Medium-term: 2–6 weeks after indexing

- Eligible pages will show FAQ dropdowns directly in Google search results
- Your search result will visually dominate the page — taking up 4–6x more space
- Users can read your answers before clicking, selecting higher-intent visitors
- Particularly impactful for /vps-server, /support, and /offers which have high purchase intent

Blog publishing system

Impact timeline: Long-term: 3–12 months of compounding returns

- Each blog post is a new URL that can rank for a unique keyword indefinitely
- Hosting tutorials ('how to install WordPress', 'what is BDIX') attract users at the research phase
- Informational content builds topical authority — Google ranks sites higher when they demonstrate expertise in a topic
- Blog posts can be linked from product pages (internal linking) to pass authority to high-converting pages
- Over 12 months of consistent publishing, organic traffic compounds significantly

Social sharing (Open Graph)

Impact timeline: Immediate: Every share

- Every link shared on Facebook, LinkedIn, Twitter, WhatsApp now shows a rich card with logo and description
- Rich previews receive significantly more engagement than plain URL text
- Consistent branding across all platforms reinforces SNBD HOST identity

Organisation schema

Impact timeline: Medium-term: 4–12 weeks

- Enables Google Knowledge Panel when users search 'SNBD HOST' directly
- Establishes the business as a verified entity in Google's knowledge graph
- Improves E-E-A-T signals — a trust factor Google uses in ranking algorithms

robots.txt

Impact timeline: Immediate: Crawl efficiency

- Prevents Google from wasting crawl budget on /admin and /api/ endpoints
- Ensures sensitive admin URLs never appear in search results
- Points all crawlers directly to the sitemap, speeding up full site discovery

SECTION 9

Deployment Guide

This section covers the steps required to deploy the full SEO + blog system on your production VPS. The website is deployed with Nginx as the reverse proxy and the blog API running as a systemd service.

Step 1: Install server dependencies

```
cd /var/www/snbdhost/server && npm install
```

Installs Express, jwt, bcryptjs, cors, sql.js, slugify in the server/ directory.

Step 2: Create server/.env

```
PORT=3001 JWT_SECRET=<64-char-random-string> ADMIN_EMAIL=admin@snbdhost.com ADMIN_PASSWORD_HASH=
CORS_ORIGIN=https://snbdhost.com
```

Generate the password hash with: node -e "console.log(require('bcryptjs').hashSync('yourpassword', 12))"

Step 3: Set up systemd service

```
/etc/systemd/system/snbd-blog-api.service WorkingDirectory=/var/www/snbdhost/server
ExecStart=/usr/bin/node index.js EnvironmentFile=/var/www/snbdhost/server/.env
```

Run: systemctl enable snbd-blog-api && systemctl start snbd-blog-api

Step 4: Configure Nginx

```
location / { try_files $uri $uri.html $uri/index.html /index.html; } location /api/ { proxy_pass
http://127.0.0.1:3001; }
```

The /api/ block proxies blog API calls to the Node.js server.

Step 5: Deploy and build

```
git pull && npm install npm run build:full
```

build:full runs vite build then scripts/generate-sitemap.cjs to include all blog post URLs.

Step 6: Submit to Google Search Console

```
Add property: https://snbdhost.com Submit: https://snbdhost.com/sitemap.xml
```

Use URL Inspection to verify Google can access and render each page.

SECTION 10

Pre-Launch Checklist

Complete the following checks before making snbdhost.com live to the public.

CRITICAL

- Change ADMIN_PASSWORD_HASH in server/.env from the development default
- Change JWT_SECRET to a 64-character random string (never commit this to git)
- Verify robots.txt accessible at https://snbdhost.com/robots.txt
- Verify sitemap.xml accessible at https://snbdhost.com/sitemap.xml
- Submit sitemap to Google Search Console
- Test admin login works at https://snbdhost.com/admin/login
- Create at least 1 published blog post before going live

SEO VALIDATION

- Use Google's Rich Results Test on /vps-server to confirm FAQPage schema validates
- Use Google's Rich Results Test on a blog post to confirm BlogPosting schema validates
- Use Facebook Sharing Debugger to verify OG tags load on /hosting
- Use Twitter Card Validator to verify card appears for /vps-server
- Check page titles are correct in browser tab for each of the 12 pages
- Verify /admin is NOT indexable (check robots.txt blocks it)

FUNCTIONAL

- Blog API health check: curl https://snbdhost.com/api/health returns {status: ok}
- Blog listing page /blog loads with posts (or empty state if no posts)
- Blog post page renders with correct title and Markdown content
- Social share buttons on blog post open correct share URLs
- Pagination works on /blog when more than 9 posts exist
- Header 'Blog' link navigates to /blog (not #)
- Admin dashboard shows all posts with correct statuses

This implementation provides a complete, production-ready SEO foundation for snbdhost.com. The combination of per-page meta tags, structured data, a dynamic sitemap, and a content publishing system gives SNBD HOST every technical advantage it needs to compete in Bangladesh's hosting market search results.

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